

Markov Chain

Discrete time,
Discrete space.

Let S be a ctble set.

Will call S state space.

Def:- A seqⁿ $\{X_n: n \geq 0\}$ is called a Markov chain taking values in S if

$$P(X_{n+1}=j \mid \underbrace{X_n=i}_{\text{present}}, \underbrace{X_{n-1}=i_{n-1}, \dots, X_0=i_0}_{\text{past}})$$

$$= P(X_{n+1}=j \mid X_n=i)$$

for all $n \geq 0$, $i, j, i_0, i_1, \dots, i_{n-1} \in S$
 whenever the prob. are well defined.
 A M.C is called homogeneous if
 $P(X_{n+1}=j \mid X_n=i)$ does not depend on n .

$$P(X_{n+1}=j | X_n=i) = P(X_1=j | X_0=i)$$

for $n \geq 1$.

Ex:- Show that for a Markov chain given the present, the past and the future are indept, i.e., for, $i, j, k \in \mathcal{S}$,

$$P(X_{n+1}=j, X_{n-1}=k | X_n=i) = P(X_{n+1}=j | X_n=i) P(X_{n-1}=k | X_n=i)$$

Define

$$p_{ij} = P(X_{n+1}=j | X_n=i) \quad S$$
$$= P(X_1=j | X_0=i), \quad i, j \in S.$$

$$0 \leq p_{ij} \leq 1.$$

$$\sum_{j \in S} p_{ij} = 1.$$

$$1 = (\underbrace{1, 1, \dots, 1}_S)$$

$P = (p_{ij})$
is called the $S \times S$
transition prob. matrix

$\underline{1}$ is
an eigen
vector
corresponding
to eigen
value 1!

$$P \underline{1}' = \underline{1}'$$

$$\sum_{j \in S} p_{ij} = \sum_{j \in S} P(X_1=j | X_0=i)$$

disj events.

$$Ax = \lambda x$$

eigen value

eigen vector.

$$\begin{pmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{pmatrix} \begin{pmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}$$

The distⁿ of X_0 is called the initial distⁿ.

Propⁿ :- Let $\{X_n : n \geq 0\}$ be a seqⁿ of r.v.s taking values in S with $X_0 \sim \mu$. Let $\{\xi_n : n \geq 1\}$ be i.i.d r.v.s taking values in T . Assume that $X_{n+1} = f(X_n, \xi_{n+1}) \forall n \geq 0$.

indep^t of X_0 .

where $f: S \times T \mapsto S$, then

$\{X_n: n \geq 0\}$ is a M.C.

Pf:- $P(X_0 = i_0, X_1 = i_1, \dots, X_k = i_k)$

$$= P(X_0 = i_0, \overset{X_1}{f(X_0, \xi_1)} = i_1, \overset{X_2}{f(X_1, \xi_2)} = i_2, \dots,$$

$$= P(X_0 = i_0, f(i_0, \xi_1) = i_1, f(i_1, \xi_2) = i_2, \dots, f(i_{k-1}, \xi_k) = i_k)$$

$$= P(X_0 = i_0) P(f(i_0, \xi_1) = i_1) \cdots P(f(i_{k-1}, \xi_k) = i_k)$$

since X_0 & $\xi_1, \xi_2, \dots, \xi_n$ are indpt

$$P(X_0 = i_0, X_1 = i_1, \dots, X_k = i_k).$$

$$= P(X_0 = i_0) \prod_{j=1}^k P(f(i_{j-1}, \xi_j) = i_j)$$

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0)$$

$$= P(f(i, \xi_{n+1}) = j)$$

$$P(X_{n+1}=j | X_n=i)$$

$$= \frac{P(X_{n+1}=j, X_n=i)}{P(X_n=i)}$$

$$X_{n+1} = f(X_n, \xi_{n+1})$$

$$= f(i, \xi_{n+1})$$

$$= \sum_{i_0, i_1, \dots, i_{n-1} \in S} P(X_{n+1}=j, X_n=i, X_{n-1}=i_{n-1}, \dots, X_0=i_0).$$

$$= P(f(i, \xi_{n+1})=j) \sum_{i_0, i_1, \dots, i_{n-1} \in S} P(X_n=i, X_{n-1}=i_{n-1}, \dots, X_0=i_0)$$

$$\begin{aligned}
 & P(X_{n+1}=j | X_n=i, X_{n-1}=i_{n-1}, \dots, X_0=i_0) \\
 &= P(X_{n+1}=j | X_n=i) = P(\underbrace{f(i, \xi_{n+1})=j}) \\
 &= P(f(i, \xi_1)=j)
 \end{aligned}$$

Hence $\{X_n : n \geq 0\}$ is a ξ_n 's are iid.
homogeneous M.C.

$$p_{ij} = P(X_1=j | X_0=i) = P(f(i, \xi_1)=j). \quad \square$$

EX 1 - $\{X_n: n \geq 0\}$ taking values in S ,
iid

then it is a M.C.

$$P(X_{n+1}=j \mid X_n=i, X_{n-1}=i_{n-1}, \dots, X_0=i_0) \\ = P(X_{n+1}=j) = P(X_{n+1}=j \mid X_n=i).$$

$$P_{ij} = P(X_{n+1}=j).$$

$$S \subset \mathbb{Z}$$

P

One row repeated.

$$\begin{pmatrix} i-1 \\ i \\ i+1 \end{pmatrix}$$

$$\begin{pmatrix} j-1 & j & j+1 \\ p_{j-1} & p_j & p_{j+1} \\ p_{j-1} & p_j & p_{j+1} \\ p_{j-1} & p_j & p_{j+1} \\ \vdots & \vdots & \vdots \end{pmatrix}$$

$$P(X_1 = j) = p_j$$

Ex 2 RW, $\epsilon_i \in \{-1, 1\}$, $P(\epsilon_i = 1) = \frac{1}{2}$
 $= P(\epsilon_i = -1)$

$$s_0 \in \mathbb{Z}. \quad S_0 = s_0, \quad S_n = s_0 + \sum_{i=1}^n \epsilon_i$$

$\{S_n : n \geq 0\}$ is a M.C

$$\epsilon_i \in \mathbb{Z}$$

$$P(\epsilon_i = k) = q_k$$

$$\mathcal{T} = \mathbb{Z}, \quad \mathcal{U} = \{-1, 1\}, \quad \mathcal{S} = \mathbb{Z}. \quad f(i, \epsilon) = i + \epsilon.$$

$$S_{n+1} = s_0 + \sum_{i=1}^{n+1} \epsilon_i = \left(s_0 + \sum_{i=1}^n \epsilon_i \right) + \epsilon_{n+1} = S_n + \epsilon_{n+1} \\ = f(S_n, \epsilon_{n+1}).$$

$$p_{ij} = P(f(i, \epsilon_1) = j) = P(i + \epsilon_1 = j) = P(\epsilon_1 = j - i)$$

$$= \begin{cases} \frac{1}{2} & \text{if } j = i+1 \text{ or } i-1 \\ 0 & \text{o.w.} \end{cases}$$

$$P = \begin{matrix} & i-1 & i & i+1 \\ \begin{matrix} i-1 \\ i \\ i+1 \end{matrix} & \begin{matrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{matrix} \end{matrix}$$

If $P(\epsilon_i = +1) = p = 1 - P(\epsilon_i = -1)$, then $\{S_n : n \geq 0\}$
is a M.C., tran prob.

$$p_{ij} = \begin{cases} p & \text{if } j = i+1 \\ q & \text{if } j = i-1 \\ 0 & \text{o.w.} \end{cases}$$

In gen, ϵ_i iid taking values in $\mathbb{Z}$
then $\mathcal{T} = \mathbb{Z}, \{S_n : n \geq 1\}$ is a M.C.

$$p_{ij} = P(\epsilon_i = j-i)$$

$$= \alpha_{j-i}$$

Higher dimensional
RW. $\{\epsilon_i : i \geq 1\}$ iid
taking values in $\mathbb{Z}^d$.
 $f(i, \epsilon) = i + \epsilon$.

$$\begin{array}{cccccc} & -2 & -1 & 0 & 1 & 2 \\ 0 & \dots & \alpha_{-1} & \alpha_0 & \alpha_1 & \alpha_2 \\ 1 & & \alpha_{-2} & \alpha_{-1} & \alpha_0 & \alpha_1 \\ 2 & & \alpha_{-3} & \alpha_{-2} & \alpha_{-1} & \alpha_0 \end{array}$$

$$\underbrace{\alpha_i \alpha_{-i+1} \dots}_{n} \quad \underbrace{\alpha_{-i} \alpha_{-i+1} \dots}_{n}$$

$$\underbrace{x_0}_{\sim} \in \mathbb{Z}^d, \quad \underbrace{S_n}_{\sim} = \underbrace{x_0}_{\sim} + \sum_{i=1}^n \epsilon_i$$

$$P_{\tilde{z}, \tilde{z}} = P(\tilde{z} + \epsilon_1 = \tilde{z})$$

$$e_i = (\underbrace{0, \dots, 0}_{i-1}, \underbrace{1, 0, \dots, 0}_{d-i})$$

$$= P(\epsilon_1 = \tilde{z} - \tilde{z})$$

Sp. Case :-

$$\epsilon_i \in \{-e_1, e_1, -e_2, e_2, \dots, -e_d, e_d\}$$

$$P(\epsilon_i = e_i) = P(\epsilon_i = -e_i) = \frac{1}{2d}$$

Simple symmetric RW
in d dim.

